

CSE 5525 Final Project Proposal

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1 Introduction

Modern LLM chat assistants are typically post-trained through multi-stage pipelines which combine supervised fine-tuning (SFT) with human preference-based alignment. In this project, we train a provided Llama-3.2-1B base model using the Tinker training infrastructure to reproduce a simplified version of this post-training pipeline. The objective is to build a correct and reproducible training pipeline that utilizes SFT and Direct Preference Optimization (DPO) to reshape the model's behavior to human preferences.

2 Methodology

We will begin by performing supervised fine-tuning on the required instruction-following dataset using parameter-efficient fine-tuning to ensure computational efficiency. This stage converts the pretrained base model into an instruction-following assistant capable of producing structured and coherent responses. The resulting model will be evaluated on the fixed General, Safety, Math, and Coding evaluation splits using the provided evaluation framework, without training on any evaluation prompts.

Starting from our best SFT checkpoint, we will then apply Direct Preference Optimization using the required preference dataset. We will compare three models: the base model, the SFT model, and the SFT + DPO model. This controlled comparison will allow us to isolate the behavioral impact of alignment and measure changes in instruction-following accuracy, safety compliance, and robustness.

3 Prior Work / Related Work

Modern post-training pipelines rely heavily on supervised tuning and preference-based alignment. InstructGPT, created by Ouyang et al., established the industry standard when they found that their fine-tuned model performed better than GPT-3

with over 100x fewer parameters (Ouyang et al.). The drawback with this model, however, is that it uses reinforcement learning from human feedback, which is highly unstable and resource intensive. This issue is addressed by utilizing direct preference optimization (Rafailov et al.). Direct preference optimization was found to be more stable, lightweight, and reproducible when creating a training pipeline (Rafailov et al.). What Rafailov et al. fail to address is what are the unintended tradeoffs created by this methodology. Our project will explicitly address this gap by documenting and quantifying these behavioral shifts when comparing the DPO-aligned model directly against a baseline SFT model.

4 Proposed Extensions

After completing the required SFT and DPO pipeline, we propose two extensions designed to deepen our understanding of fine tuning and alignment tradeoffs.

4.1 Data Filtering and Quality Analysis

For extension one, we will perform data filtering and quality analysis on the instruction and preference datasets. The dataset will be analyzed at first to understand its composition, examining properties such as response length distributions, duplicate and near-duplicate frequency, and stylistic variations. Based on this analysis, we will construct filtered subsets of data using several possible approaches: length filtering to remove outlier responses, deduplication, and quality-based post training heuristics such as filtering out high-perplexity responses under the base model. For each filter, we will form a hypothesis about its expected effect on model behavior before training. We will then train models on both the full and filtered datasets and compare their behavior across all four evaluation splits to study how data quality and

composition influence both supervised fine-tuning and alignment outcomes. We will select two to three well-justified filters rather than exhaustively testing all possible combinations due to possible resource constraints.

4.2 Safety vs Helpfulness Tradeoff Curves

The second extension will quantify the tradeoff of safety versus helpfulness through testing variations in the alignment strength parameter during DPO training. For each variation, the rate of refusal on safety prompts and task performance on general benchmarks will be tracked and plotted. This will allow us to observe how increasing alignment pressure affects the delicate balance between safety and helpfulness.

All experiments will be conducted with credit budget and reproducibility in mind. By documenting configurations, hyperparameters, random seeds, and evaluation outputs, we can ensure the results will be easily replicated and compared.

References

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